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position. The heating is continued till the brown color of the liquid first produced, disappears leaving the contents clear as before. At this point all the nitrogen in the substance is converted to $(NH_4)_2SO_4$. The Kjeldahl flask is then cooled and its contents are diluted with same distilled water and then carefully transferred into a 1 lit round bottom flask. An excess NaOH solution is poured down the side of the flask and it is fitted with a Kjeldahl trap and a water condenser. The lower end of the condenser dips in a measured volume of excess the $\frac{N}{20}$ H_2SO_4 solution. The liquid in the round bottom flask is then heated and the liberated ammonia is distilled into sulphuric acid. The Kjeldahl trap serves to retain any alkali splashed up on vigorous boiling. (Refer Page 153 Fig. 11.3)

When no more ammonia passes over (test the distillate with red litmus) the receiver is removed. The excess of acid is then determined by titration with alkali, using phenolphthalein as the indicator.

Calculation:

Weight of the substance = Wg.

Volume of H_2SO_4 required for the complete neutralisation of evolved NH_3 = V mL.
Strength of H_2SO_4 used to neutralise NH_3 = N

Let the Volume and the strength of NH_3 formed are V_1 and N_1 respectively

we know that $V_1N_1 = VN$

The amount of nitrogen present in the w g of Organic Compound = $\frac{14 \times NV}{1 \times 1000 \times w}$

Percentage of Nitrogen = $\left(\frac{14 \times NV}{1000 \times w}\right) \times 100\%$

Example : 0.6 g of an organic compound was Kjeldalised and NH_3 evolved was absorbed into 50 mL of semi-normal solution of H_2SO_4. The residual acid solution was diluted with distilled water and the volume made up to 150 mL. 20 mL of this diluted solution required 35 mL of $\frac{N}{20}$ NaOH solution for complete neutralization. Calculate the % of N in the compound.

Weight of Organic compound = 0.6g
Volume of sulphuric acid taken= 50mL
Strength of sulphuric acid taken= 0.5 N
20 ml of diluted solution of unreacted sulphuric acid was neutralised by 35 mL of 0.05 N Sodium hydroxide

Strength of the diluted sulphuric acid = $\frac{35 \times 0.05}{20}$

= 0.0875 N

Volume of the sulphuric acid remaining after reaction with Organic compound = V_1 mL
Strength of H_2SO_4 = 0.5N
Volume of the diluted H_2SO_4 = 150 mL
Strength of the diluted sulphuric acid = 0.0875 N

$V_1 = \frac{150 \times 0.087}{0.5}$ = 26.25 mL

Volume of H_2SO_4 consumed by ammonia = 50 - 26.25

= 23.75 mL

23.75 mL of 0.5 N $H_2SO_4 \equiv$ 23.75mL of 0.5N NH_3

The amount of Nitrogen present in the 0.6 = $\frac{14g}{1000\ mL \times 1\ N} \times 23.75 \times 0.5N$

= 0.166g